

Tahawar Ihsan

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Experience

Software Engineer (AI Core Team)

Aug 2024 – Present

NetSol Technologies Ltd. Lahore, Pakistan

- Spearheaded the AI module for **Check** and **CDE (Credit Decision Engine)**, enabling human-in-the-loop underwriting with scalable, compliant, and explainable agentic AI workflows.
- Designed and implemented an **agentic chatbot system** for CDE using **LangGraph**, **LiteLLM** (for key management, model routing, and cost control), and **MCP ContextForge** for safe, structured tool-calling across business logic services.
- Built event-driven orchestration pipelines by integrating **n8n** to trigger agent workflows, automate decision stages, and streamline multi-step credit evaluation processes.
- Architected agentic microservices using **RabbitMQ** and **Celery**, supporting semantic retrieval, underwriting Q&A, and text-to-SQL capabilities over enterprise datasets.
- Worked on multi-tenant architecture for **Check**, enabling secure tenant-based login, configuration, and data separation so multiple clients can use the same platform safely.
- Worked on **Flow** for document extraction to ingest partner documents and process monthly rate charts, then publish validated results to production systems.
- Built a POC real-time voice agent using **LiveKit** (WebRTC), with **Vapi** orchestration, **Deepgram** streaming ASR, and **ElevenLabs** low-latency TTS.
- Used **AWS Textract** with validation and post-processing to extract rate chart fields such as residual value and lease terms, calculate required values, and publish final outputs to production.
- Built bank statement analysis workflows to identify transaction patterns and spending behavior, generating structured insights to support risk and affordability decisions.
- Improved observability of LLM-based services with **Langfuse**, **DSPy**, and **Logfire**, enabling compliance-ready auditing and real-time performance monitoring.
- Developed NLP-driven data retrieval systems for **BMW** and **Mercedes** using **MongoDB Atlas Vector Search** and **Pinecone** to optimize insight discovery.
- Built robust, scalable AI APIs with **FastAPI** and **Docker**, integrating **OpenAI**, **Claude**, and **Llama** models for enterprise use cases.

Machine Learning and Python Engineer

Jul 2023 – Jul 2024

Codrivty UK, Islamabad

- Led development of an AI-driven recruitment system (**OnboardIQ**), achieving 95% accuracy in candidate screening.
- Built Lumina Chatbot (Django-based), enhancing user engagement in an Agile environment.
- Developed algorithm to auto-generate interview questions from extracted keywords, reducing HR prep time by 40%.
- Contributed to API development and documentation (Postman, Swagger UI, Redocly) for seamless integration.
- Deployed solutions on **AWS Lambda**, leveraging **ECS** and **S3** for scalable, reliable performance.

Languages and Technologies

- **Core:** Python, SQL, JavaScript
- **Backend:** FastAPI, Django, REST APIs
- **Agentic/LLM:** LangGraph, LangChain, LiteLLM
- **Machine Learning & AI:** TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas
- **Vector & Retrieval:** Pinecone, MongoDB Atlas Vector Search
- **Data:** PostgreSQL, MongoDB, MySQL
- **Messaging/Orchestration:** RabbitMQ, Celery, n8n
- **Cloud/DevOps:** Docker, AWS (Lambda, ECS, S3, Bedrock), Azure DevOps
- **Testing/Observability:** Pytest, Postman, Locust, Langfuse, Logfire

Projects

- **Secure ML Pipeline with Homomorphic Encryption (FYP):** Implemented privacy-preserving ML models using homomorphic encryption.
- **Attendance Analysis of Pakistan's National Assembly Members:** [Link](#)
- **LangChain-based Chatbot:** Built a chatbot powered by LangChain to query CSV files.
- **Heart Disease Prediction:** [Link](#)
- **Time Series Forecasting with TensorFlow:** [Link](#)

Education

National University of Computer and Emerging Sciences, FAST-NUCES

Lahore, Pakistan

Bachelor of Science in Computer Science

CGPA: 3.02

Certificates

- **Machine Learning Specialization (Coursera)** ([link](#))
Courses: Supervised ML, Advanced Learning Algorithms, Unsupervised Learning & Recommenders